

I. Convert “Project” → “Research Problem”

- **Current state:**

Your work is descriptive:

“you have built a system with AI detection, alerts, etc.”

- 1) Required transformation:**

You must **frame a research gap**

- a) Define the problem clearly:**

Example:

Existing smart surveillance systems rely heavily on cloud infrastructure, leading to privacy risks, latency issues, and dependency on continuous internet connectivity.

- b) Your research objective:**

To design and evaluate a **privacy-preserving, fully edge-based intelligent surveillance system** with real-time detection and secure local processing.

II. Strengthen the Technical Architecture Section

Your report describes features—but journals need **structured architecture**.

A) Add:

(a) System Block Diagram

Include:

- Camera input
- AI detection module (YOLOv8)
- Alert engine
- Authentication module
- Database (SQLite)
- UI (Web + macOS app)

(b) Data Flow Explanation

Explain step-by-step:

1. Frame captured

2. Preprocessing
3. Object detection
4. Classification (normal/danger)
5. Decision logic
6. Alert generation
7. Logging

This must be written **algorithmically**, not descriptively.

III. Add Mathematical / Algorithmic Modelling

A) Add models for:

(a) Detection Confidence

$$P_{detect} = \frac{TP}{TP + FN}$$

(b) False Alarm Rate

$$FAR = \frac{FP}{FP + TN}$$

(c) System Latency

$$T_{total} = T_{capture} + T_{processing} + T_{alert}$$

IV. MOST IMPORTANT: Experimental Validation

a) Dataset Creation

You must create your own dataset:

- Indoor/outdoor scenarios
- Day/night
- With/without human presence
- With danger objects (knife, scissors, etc.)

b) Performance Metrics

Evaluate:

Metric	Meaning
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Accuracy	Detection correctness
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Precision	False alarms control
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Recall	Missed detections
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F1-score	Balanced metric
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FPS	Real-time capability
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Latency	System delay
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c) Example Results Table

Scenario	Accuracy	FPS	Latency
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Daytime	x%	xx	xx ms
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Night	Y%	xx	Xx ms
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d) Comparative Study

Compare your system with:

1. Cloud-based systems

- Higher latency
- Privacy risk

2. Other edge-based systems (from literature)

Highlight:

- **Zero cloud dependency (your work)**
- Lower latency
- Better privacy

5. Add Security Analysis Section

Include:

Threat Model

- Unauthorized access
- Password attacks
- Network interception

Defense Mechanisms

- OTP authentication
- Password hashing using bcrypt
- Rate limiting

Advanced addition (high impact):

- Session timeout
- Token rotation
- Login logout

6. Add Intelligence Layer (Novelty Boost)

Right now:

- Detection → Alert

Upgrade to:

Context-Aware Decision System

Example logic:

- If person detected + night + no registered device → HIGH alert
- If known device present → suppress alert

This becomes a **decision model**, not just detection

7. Add Ablation Study

Test system **with and without features**

Feature	Accuracy	False Alerts
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Without presence detection	%	
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With presence detection	%	
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8. Performance Optimization Study

Take YOLOv8 and:

- Apply lightweight version / pruning
- Compare:

Model	FPS	Accuracy
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Original		%
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Optimized		%
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9. Literature Review (Strong Section)

Include:

- Edge AI surveillance systems
- Privacy-preserving AI
- Smart home security

Then clearly state:

“Unlike existing systems, Garuda performs complete processing locally without any cloud dependency.”

10. Add Diagrams (MANDATORY)

You need:

- System architecture diagram
- Data flow diagram
- Detection pipeline
- Alert mechanism flowchart

11. Highlight Your Novel Contributions Clearly

At the end of introduction:

Example:

- Fully local AI-based surveillance system
- Privacy-preserving architecture
- Context-aware alert mechanism
- Integrated multi-modal interface (web + app + voice)